Pricing Strategy and Retail Competition at Dollar Stores

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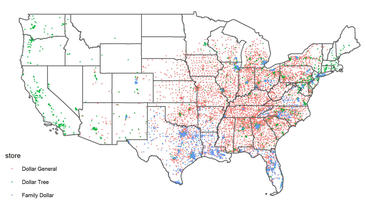
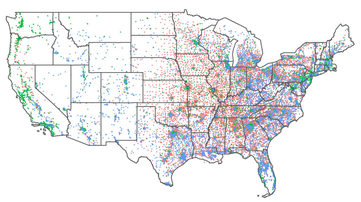
Motivation

- Rapid expansion of dollar stores across the country over the past two decades
- Dollar stores offer unprecedented level of affordability, but may also replace access to fresh fruit, vegetables and meat with lower-priced processed foods

2008



2019



Research Question and Data

Research Questions:

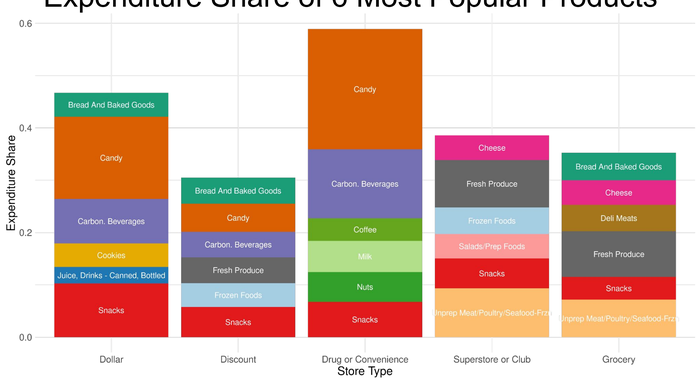
1. What is a dollar store? What do dollar stores sell?
2. Who shops at the dollar store?
3. Compare pricing strategies across chains and other retailers

Data:

1. Kilts-Nielsen Data: panel of 40,000-60,000 US households and their purchases by store, 2007-2019
2. SNAP Historical Retailers: location, opening, closing dates of dollar stores and grocery stores as they enter in the SNAP sample

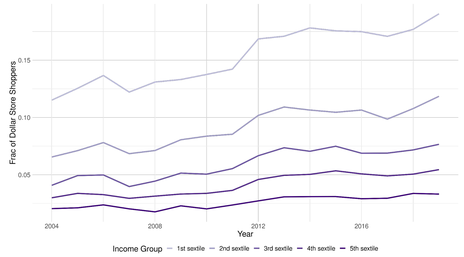
Dollar Store Products

Expenditure Share of 6 Most Popular Products



Consumer Base

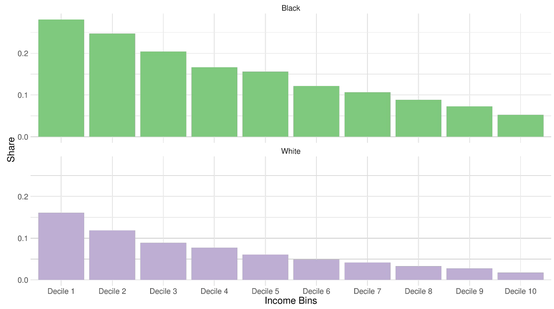
Growth of Dollar Store Shoppers by Income Group



- An increasing fraction of lower income consumers have become dollar store shoppers over time
- Black households have significantly higher shares of dollar store shoppers in every income bracket relative to white households

Share of income group who are dollar store and discount store shoppers

- Note: Dollar store shopper is defined as household who spent more than 5 percent of their expenditures at a dollar store, but these general trends hold with different cutoffs



Average Price Differences

Dollar stores are able to offer lower prices, on average, by selling cheaper brands and smaller sizes within product groups

$$log(y_{it}) = StoreType + \alpha_{ict} + \epsilon_{ict}$$

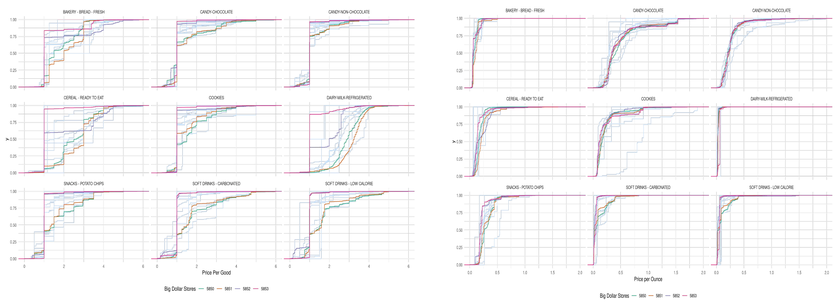
- y(i,t) is the price of good i in month-year t
- StoreType: dollar or discount store, reference group: grocery stores, club/superstores, convenience stores
- Include good x product x month fixed effect

Price Variables:	log(price_per_good)	log(price_per_good)	log(price_per_unit)
Variables			
DollarStore	-0.1312	-0.5155	-0.2581
	(0.0141)	(0.0129)	(0.0091)
Discount	-0.0494	-0.0277	-0.0410
	(0.0017)	(0.0029)	(0.0020)
Fixed-effects			
county_upc_month	Yes		
county_product_month		Yes	Yes
Fit statistics			
Observations	603,875,580	603,875,580	603,875,580
R^2	0.87564	0.45083	0.80180

Clustered (county_name) standard-errors in parentheses

Price Differences Across Dollar Stores

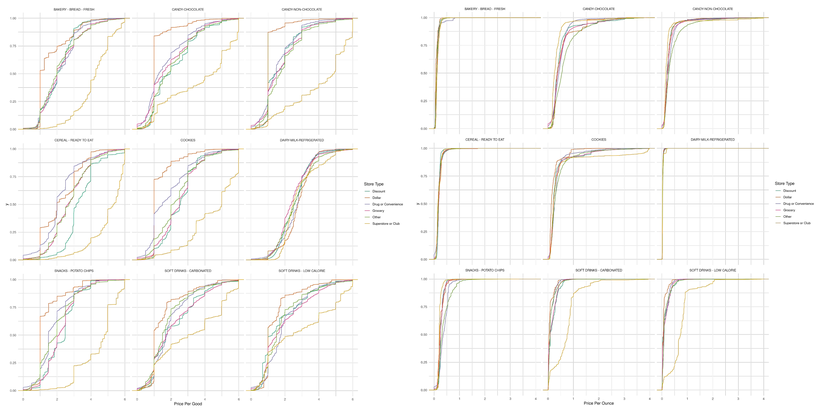
CDF of Prices Across Different Dollar Store Chains



- Dollar store consistently employ a $1 pricing scheme. However, there is some heterogeneity in this price constraint, which can be seen by plotting the CDFs of prices and price per unit for various products across chains.

Price Differences Across Retailers

CDF of Prices Across Retailers



- Dollar stores usually offer a lower distribution of prices relative to other retailers. However, their ability to provide discounts relative to other stores varies substantially across products.

Acknowledgements

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